

Placing Tourist Spots in Smart Cities: A Crowdsourced Game Theoretic Approach

Surja Sanyal 

Department of Computer Science and Engineering
Indian Institute of Technology Guwahati
Guwahati, Assam, India
surja.sanyal@iitg.ac.in; surja.sanyal@iitbhu.ac.in

Tanu Priya

Department of Computer Science and Engineering
National Institute of Technology Durgapur
Durgapur, West Bengal, India
tanupriya26.1999@gmail.com

Naren Debnath 

Department of Computer Science and Engineering
Narula Institute of Technology
Kolkata, West Bengal, India
naren.debnath@nit.ac.in

Debasis Mitra , Sajal Mukhopadhyay 

Department of Computer Science and Engineering
National Institute of Technology Durgapur
Durgapur, West Bengal, India
dmitra.cse@nitdgp.ac.in; smukhopadhyay.cse@nitdgp.ac.in

Abstract—Local government frequently undertakes new projects to encourage the tourism industry. Selecting a tourist spot by aggregating stakeholder preferences is challenging, and involving local citizens through crowdsourcing makes the selection process transparent and engaging. Existing solutions have not considered the strategic setting in which single-peaked preferences are to be provided, namely, two or more pairwise parallel roads in the context of selecting a probable facility construction point (tourist spot) enclosed by these roads. Here, the aim is to aggregate the preferences of residents along the perimeter of this available space to select the spot. The proposed solution uses game theory to address this challenge, leveraging the technological platforms available in smart cities and multidimensional preference projection techniques. The theoretical results are validated with the simulation results by carefully crafting the benchmark mechanism.

Index Terms—Tourist Spot Selection, Voting, Game Theory, Participatory Democracy, Mobile Crowdsourcing, Single-Peaked Preference, Multidimensional Preference, Preference Projection.

I. INTRODUCTION

Governments worldwide have recognized the tourism industry as a major source of income for their countries and have continually undertaken new projects to provide better facilities to encourage its growth. Previously, decisions on selecting a project site were made solely by the government. Recently, the residents of the land have also been involved in the same [1], [2]. This results in greater transparency and better resident engagement in the selection process [3]–[7]. This also favorably modifies residents' attitudes towards the industry [8], [9]. The availability of a variety of technological platforms in the existing and upcoming smart cities has made crowd involvement in decision-making and problem-solving even more feasible, and the mass percolation of such solutions possible through digital advancements in the areas of healthcare, education, transport, communication, and resource management, among others [10]. These platforms, available in smart cities, are used by various groups, such

as governments, businesses, and researchers, to engage more citizens in decision-making, design better products for the masses, and develop better solutions [11]. The residents' attitude towards tourism has been studied and discussed in detail using various papers in [1]. However, these studies focus more on individual case studies rather than on a community. This portrays the importance of involving residents in the sustainable development of the tourism sector [3]–[7].

When agents are engaged in the decision-making process, it may be natural to assume they are strategic. Aggregating their preferences into a single joint decision is no longer a trivial problem (when we need that, the mechanism should be strategyproof). The multidimensional variation of the single-peaked voting rule is strategyproof [12]. Real-life voting using the concept of participatory democracy engages the stakeholders or the residents of the land by considering their preferences in selecting the desired outcomes [13], [14]. Crowdsourcing applications, including data labeling, geographic data collection, efficient tourist guidance systems, service demand prediction, indoor positioning systems, and intelligent transportation, have gained popularity [5], [15]–[17]. With the support of technology, tourism experiences are becoming better and more competitive for smart tourism project development [18], [19]. Mobile crowdsourcing as a convenient way to recruit agents has used methods that conditionally recruit based on their capabilities [20].

Participatory democracy using single-peaked preferences and the median voting rule has been used, in a smart city context, to select tourist spots in a city with the help of city dwellers [21]. The median voting rule has been proven to be strategyproof in aggregating the votes collected using available technology in the city and on the Internet. In [21], a tourist spot in a smart city context was selected in a crowdsourced manner with the involvement of the city dwellers present in the locality by considering that the residents are located on one side of a long road (the length of the long road is considered

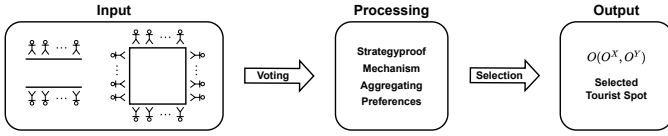


Fig. 1. Overall representation of the proposed solution for spot selection.

unity) and the residents can vote for the tourist spot within the $[0, 1]$ interval of that road. However, in the scenario where the residents are lying along two parallel roads (as shown in Fig. 2(a)) or along the border of a rectangle (as shown in Fig. 2(b)) wherein they need to select the tourist spot in a suitable place in the region considered (as shown in Fig. 2(a) and Fig. 2(b)), is not addressed in [21].

In this work, we consider a smart city scenario in which single-peaked preferences are submitted from two parallel sides of an enclosed space for facility construction. We also consider the scenario of two pairs of pairwise parallel roads enclosing the available space on all four sides, and the residents submit their votes as single peaks in a two-dimensional voting space. We modify the one-dimensional applicability of the single-peaked preferences and the median voting rule mentioned in [21] in a multidimensional scenario to fit our aforementioned settings, motivated by [12], [22], [23]. The overall representation of the problem statement and the proposed solution is depicted in Fig. 1.

The following is the organization of this paper. The problem has been formalized, and the proposed mechanism is described in Section II. The proposed mechanism is analyzed in Section III, followed by its simulation results in Section IV. Discussion on the results is presented in Section V. Conclusion and future work are put forward in Section VI.

II. ALGORITHMS AND METHODOLOGY

Here, the problem is formalized, and a solution is proposed.

A. System Model

In our model, we have one facility to be placed in between the two or more pairwise parallel roads $R \in \{A, B, C, D\}$, where $A \parallel B$ and $C \parallel D$. We treat voters and their preferences as two distinct entities. The voters can be represented as a set $\vartheta = \{\vartheta_1, \vartheta_2, \dots, \vartheta_n\}$. The voters could further be divided into four sets, such that the voters of each set can be represented by $\vartheta_R = \{\vartheta_{R_1}, \vartheta_{R_2}, \dots, \vartheta_{R_{|\vartheta_R|}}\}$, according to their locations of residence being beside roads $R \in \{A, B, C, D\}$. For example, say $R = A$, then $\vartheta_{\{R=A\}} = \{\vartheta_{A_1}, \vartheta_{A_2}, \dots, \vartheta_{A_{|\vartheta_A|}}\}$. The number of voters in each set are $|\vartheta_R| = n_R$. So, $\sum_R |\vartheta_R| = |\vartheta| = n$. Through voting, the strategic residents attempt to reach a collective decision on where the facility will be placed between the two pairs of mutually parallel roads. We can define the set $\alpha_{pref_R} = \{l_{R_1}, l_{R_2}, \dots, l_{R_{n_R}}\}$, where $R \in \{A, B, C, D\}$. That is, for $R = A$, $\alpha_{pref_{\{R=A\}}} = \{l_{A_1}, l_{A_2}, \dots, l_{A_{n_A}}\}$. Each l_{R_i} , given by ϑ_{R_i} , is a tuple with two attributes $(l_{R_i}^X, l_{R_i}^Y)$, where each of

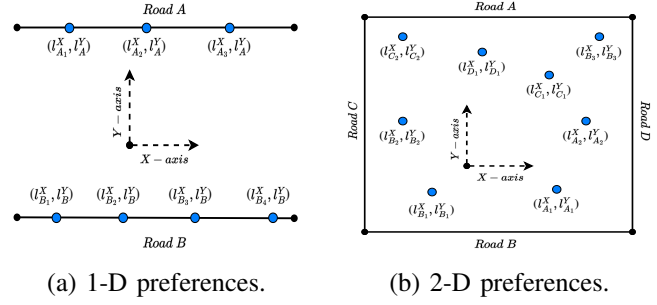


Fig. 2. Single-peaked preferences submitted on sets of mutually parallel roads.

the two attributes lies within an interval $[0, 1]$, i.e., $l_{R_i}^X \in [0, 1]$ and $l_{R_i}^Y \in [0, 1]$.

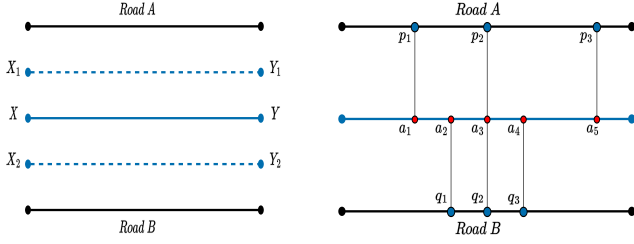
Each voter can give only one single-peaked preference, either on the X-axis for 1-dimensional voting or on both axes for 2-dimensional voting. So, each l_{R_i} corresponds to a location (preference) in a 2-dimensional space as shown in Fig. 2(a) for the 1-dimensional voting, and Fig. 2(b) for the 2-dimensional voting. Same location preferences submitted by two voters are counted as two different votes, i.e., for two voters ϑ_{R_j} and $\vartheta_{R_{j'}}$, we have $(l_{R_j}^X, l_{R_j}^Y) \neq (l_{R_{j'}}^X, l_{R_{j'}}^Y)$ even if $l_{R_j}^X = l_{R_{j'}}^X$ and $l_{R_j}^Y = l_{R_{j'}}^Y$. We need to select a position $O(O^X, O^Y)$ for the facility, taking into account residents' preferences. As residents will provide these preferences via mobile devices, this system can be considered a mobile crowdsourcing solution, leveraging the technologies available in existing and upcoming smart cities to address the problem at hand.

For the setting using two parallel roads as in Fig. 2(a), the submitted preferences take the form $(l_{R_i}^X, l_{R_i}^Y)$ for the voter ϑ_{R_i} , who resides beside the road $R \in \{A, B\}$ with $A \parallel B$, where $i \in [0, n_R]$ and $l_{R_i}^Y$ is the y-coordinate for the residents beside the road R . Here, voters are allowed to give preferences only along the X-axis. Voters should give their preference of location in the same road they reside on, i.e., $(l_{R_i}^X, l_{R_i}^Y)$ for voter ϑ_{R_i} who resides beside road R . For the setting using two pairs of mutually parallel roads as in Fig. 2(b), the allowed preferences take the form $(l_{R_i}^X, l_{R_i}^Y)$ for the voter ϑ_{R_i} , who resides beside the road $R \in \{A, B, C, D\}$ with $A \parallel B$ and $C \parallel D$, where $i \in [0, n_R]$. Voters give their single-peaked preferences in the 2-dimensional space enclosed by the two sets of mutually parallel roads in R .

B. Proposed Mechanism

We discuss two scenarios for voting. One in which the available space is enclosed within two parallel roads. The other has two pairs of mutually parallel roads enclosing the available space.

1) *Voting with two parallel roads*: This mechanism comes into play when there are two parallel roads and a need to construct a facility in the space between them, as shown in Fig. 2(a). Strategic voters on both sides submit their preferences, and our objective is to design a strategyproof mechanism that is also optimal according to our definition of



(a) Selection of a middle line. (b) Projection of preferences.

Fig. 3. Selection of a middle line and projection of the preferences onto the same for voting with single-peaked preferences along one dimension.

Algorithm 1: X-Coordinate Mean for 1-D Voting.

Input: A set of points

$$(l_{R_1}^X, l_{R_1}^Y), (l_{R_2}^X, l_{R_2}^Y), \dots, (l_{R_n}^X, l_{R_n}^Y)$$

Output: The selected location on the middle line using the mean approach.

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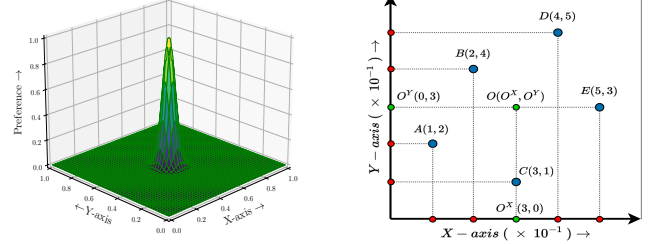
1  $sum \leftarrow 0$ 
2 for  $i \leftarrow 1$  to  $n$  do  $sum \leftarrow sum + l_{R_i}^X$ 
3  $mean \leftarrow sum/n$ 
4  $middle \leftarrow (l_{R=A}^Y + l_{R=B}^Y)/2$ 
5 return  $mean, middle$ 

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optimality. Let us assume that the first road lies on the X-axis, and the other road is parallel to it. Here, the first step is to choose a line between the two parallel roads. We can choose a parallel line halfway between the two roads so that people on both sides don't argue, as shown in Fig. 3(a).

Definition 1. *Optimality is defined as the maximization of the collective utility $U_R^Y = \sum_{R \in \{A, B\}} \sum_{i \in [1, n_R]} u_{R_i}^Y$ for all agents along the Y-axis taking each road $R \in \{A, B\}$ individually. Here, individual utility along the y-axis is defined as $u_{R_i}^Y = 1 - |l_{R_i}^Y - O^Y|$ where $O(O^X, O^Y)$ is the selected point for the facility.*

As shown in Fig. 3(a), it is observed that the utility of an agent along the Y-axis depends on the Y-coordinate of the line chosen parallel to the roads. The closer the chosen line to the agent, the higher the utility of that agent along the Y-axis. Taking the middle line as X_1Y_1 increases $U_{R=A}^Y$ and reduces $U_{R=B}^Y$. Similarly, taking the middle line as X_2Y_2 reduces $U_{R=A}^Y$ and increases $U_{R=B}^Y$. Thus, the middle line is taken as XY , thereby maximizing both $U_{R=A}^Y$ and $U_{R=B}^Y$. We project all points from both sides on the middle line as shown in Fig. 3(b). Thereafter, we may choose either method to calculate the optimal point. One is the mean method, and the other is the median method. We present the mean method in Algorithm 1. Here, we calculate O^Y as the Y-coordinate of the middle line. Then, we calculate O^X as the *mean* of all the preferences projected on the middle line. The median method, on the other hand, is strategyproof and is presented in Algorithm 2. Here, we calculate O^Y as the Y-coordinate of



(a) A sample 2-D preference. (b) Preference projections.

Fig. 4. Two-dimensional single-peaked preferences and their projections.

Algorithm 2: X-Coordinate Median for 1-D Voting.

Input: A set of points

$$(l_{R_1}^X, l_{R_1}^Y), (l_{R_2}^X, l_{R_2}^Y), \dots, (l_{R_n}^X, l_{R_n}^Y)$$

Output: The selected location on the middle line using the median approach.

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1  $X = \{l_{R_1}^X, l_{R_2}^X, \dots, l_{R_n}^X\}$ 
2  $X \leftarrow \text{Sort}(X)$ 
3 if  $n \% 2 \neq 0$  then  $count \leftarrow n + 1$ 
4 else  $count \leftarrow n$ 
5  $median \leftarrow l_{R_{(count/2)}}^X$ 
6  $middle \leftarrow (l_{R=A}^Y + l_{R=B}^Y)/2$ 
7 return  $median, middle$ 

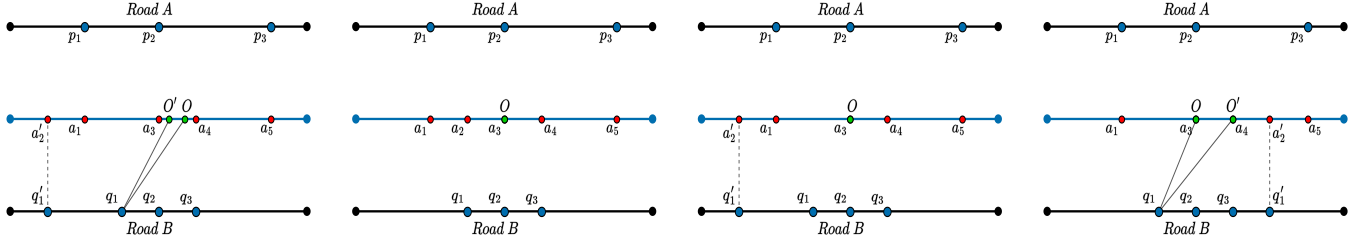
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the middle line. Then, we calculate O^X as the *median* of all the preferences projected on the middle line. Thus, we obtain the selected point $O(O^X, O^Y)$.

2) *Voting with two pairs of mutually parallel roads:* For the next scenario, consider the facility's construction area to be enclosed within two pairs of mutually parallel roads, as in Fig. 2(b). Here, voters from all four roads submit their individual preferences (l_{R_i}) as shown in Fig. 4(a). We need to provide a strategyproof mechanism to select the facility's location. We again present two approaches for the mechanism, with their projections on both axes shown in Fig. 4(b). Say, the original location (l_{R_i}) is $(0.4, 0.5)$, Fig. 4(b) shows the same in the scaled-up notation for clarity as $(4, 5)$ where the multiplier is $\times 10^{-1}$ to obtain the original location. The mean method is presented in Algorithm 3, where we select the point $O(O^X, O^Y)$ as the arithmetic mean of the projections of the preferences on both axes. However, this method is not strategyproof, as we will see later. The median method, presented in Algorithm 4, selects the point that is the median of the projections of the preferences onto both axes. This method proves to be strategyproof. Both methods are analyzed in Section III with respect to their emergent properties.

III. ALGORITHM ANALYSIS

In this section, we will analyze the strategyproofness of the proposed methods. We define the individual utility of the i^{th}



(a) Mean method with misreport. (b) Median method without misreport. (c) Median unchanged with misreport. (d) Median changed with misreport.

Fig. 5. Analysis of the movement of the selected point $O(O^X, O^Y)$ using the mean and the median methods as an agent misreports.

Algorithm 3: Both Coordinate Means for 2-D Voting.

Input: A set of points

$$(l_{R_1}^X, l_{R_1}^Y), (l_{R_2}^X, l_{R_2}^Y), \dots, (l_{R_n}^X, l_{R_n}^Y)$$

Output: The mean of the locations on both axes.

1 $sum_x, sum_y \leftarrow 0, 0$

2 **for** $i \leftarrow 1$ **to** n **do**

3 $sum_x \leftarrow sum_x + l_{R_i}^X$
4 $sum_y \leftarrow sum_y + l_{R_i}^Y$

5 $mean_x \leftarrow sum_x / n$

6 $mean_y \leftarrow sum_y / n$

7 **return** $mean_x, mean_y$

Algorithm 4: Both Coordinate Medians for 2-D Voting.

Input: A set of points

$$(l_{R_1}^X, l_{R_1}^Y), (l_{R_2}^X, l_{R_2}^Y), \dots, (l_{R_n}^X, l_{R_n}^Y)$$

Output: The median of the locations on both axes.

1 $X = \{l_{R_1}^X, l_{R_2}^X, \dots, l_{R_n}^X\}$

2 $Y = \{l_{R_1}^Y, l_{R_2}^Y, \dots, l_{R_n}^Y\}$

3 $X, Y \leftarrow \text{Sort}(X), \text{Sort}(Y)$

4 **if** $n \% 2 \neq 0$ **then** $count \leftarrow n + 1$

5 **else** $count \leftarrow n$

6 $median_x \leftarrow l_{R_{(count/2)}}^X$

7 $median_y \leftarrow l_{R_{(count/2)}}^Y$

8 **return** $median_x, median_y$

resident beside the R^{th} road (agent ϑ_{R_i}) as:

$$u_{R_i} = u_{R_i}^X + u_{R_i}^Y, \quad (1)$$

where $u_{R_i}^X = 1 - \left| l_{R_i}^X - O^X \right|$,
and $u_{R_i}^Y = 1 - \left| l_{R_i}^Y - O^Y \right|$.

Lemma 1. *Voting with single-peaked preferences on parallel lines using the mean method is not strategyproof.*

Proof. In the 1-dimensional voting, where the available space is between two parallel roads, using the mean method yields the X-coordinate of the chosen point O as $O^X = \frac{1}{n_A + n_B} \times \sum_{R \in \{A, B\}} \sum_{i=1}^{n_R} l_{R_i}^X$. In Fig. 5(a), where the preferences are denoted as $p_1, p_2, \dots, p_{n_A}$ and $q_1, q_2, \dots, q_{n_B}$ for the two roads, respectively. The projections of the preferences onto the middle line are denoted as $a_1, a_2, \dots, a_{n_A + n_B}$. Preference q_1 is projected to a_2 . If the agent providing preference q_1 misreports as q'_1 . The new mean is at point O' . If $O^X > a_2$ and $q'_1 < q_1$, which implies $a'_2 < a_2$, then $O'^X < O^X$. Hence, point q_1 is much closer to O' than O , making $u_{q'_1}^X > u_{q_1}^X$. Thus, the mean method fails to be strategyproof. Similarly, in the 2-D voting, where the mean method is applied to the projections of the preferences onto both axes, it is not strategyproof. $\square$

Lemma 2. *Voting with single-peaked preferences on parallel lines using a median rule is strategyproof.*

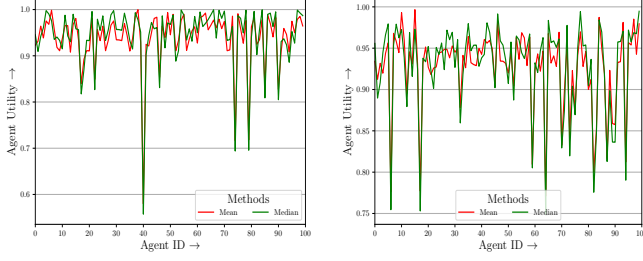
Proof. In Fig. 5(b), if all agents submit preferences truthfully, then the median is chosen as $O(O^X, O^Y)$ where a_3 . In Fig. 5(c), the agent with the preference at q_1 misreports and gives his preference at q'_1 such that q'_1 is projected to a'_2 on the middle line. Since $a_2 < O^X$ and $a'_2 < O^X$, there will be no change in the median and the utility of the agent. But in Fig. 5(d), if q_1 is misreported at q'_1 , then its projection on the middle line is at a'_2 . Here, $a_2 < O^X$, but $a'_2 > O^X$ and the new median is at a_4 at point O' further away from the misreporting agent. In this case, the utility of the agent has decreased as $u_{q'_1}^X < u_{q_1}^X$. Thus, there is no incentive to misreport under the median method. Similarly, it can be argued for the 2-dimensional preferences where the median method is applied to the projections of the preferences onto the axes. $\square$

IV. RESULTS

This section discusses the simulation setup and analyzes the results for the proposed and state-of-the-art mechanisms.

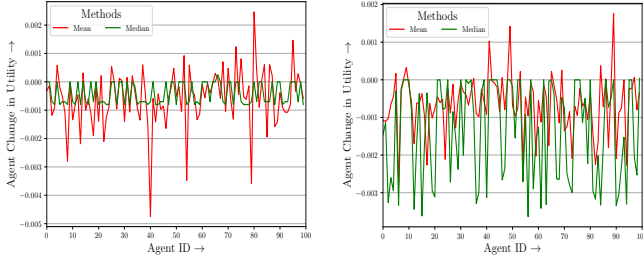
A. Setup

Real data was taken from OPEN DATA BARCELONA and cast to the simulation settings using Python libraries. Most simulations were run with 100 agents to simplify the graphs. Wherever applicable, the total number of agents was varied from 100 to 1000 in steps of 100. An absolute agent utility



(a) 1-Dimensional voting. (b) 2-Dimensional voting.

Fig. 6. Agent utilities for each agent while all agents report honestly, using 100 agents with mean and median voting rules.



(a) 1-Dimensional voting. (b) 2-Dimensional voting.

Fig. 7. Change in utilities for dishonest voting by one agent while the other agents report honestly, using 100 agents with mean and median voting rules.

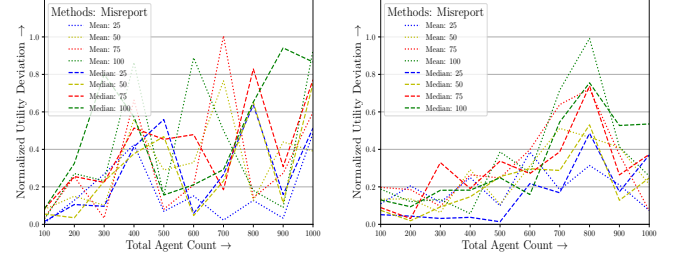
of 1 was assumed to be the maximum possible value for the same, which results if the preferred point provided by an agent coincides with the one selected by the system. However, utility changes are measured from a baseline of 0, which implies no change at all. The percentage of misreporting varied across the values $\{25, 50, 75, 100\}$, with the remaining agents reporting honestly. The absolute distance between the truthful and the misreported preferences has been taken as the measure of an agent's deviation. For the state-of-the-art, average voting rule [22], the proposed method is compared.

B. Analysis

Here, Fig. 6 shows that the utility for agents when everyone reports honestly is at most 1 using both the mean and the median methods for both voting scenarios. However, when one of the agents misreports, as in Fig. 7, the utility of that specific agent may change to greater than or less than the utility achieved using an honest preference for both the 1- and 2-dimensional voting scenarios using the mean method. The median method for both voting scenarios ensures the utility is at most what it would be under honest preferences, even when the agent misreports. Given, the utility of agent ϑ_{R_i} is:

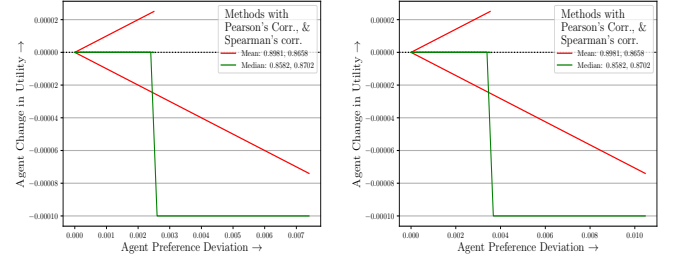
$$u_{R_i} = \left[1 - |l_{R_i}^X - O^X| \right] + \left[1 - |l_{R_i}^Y - O^Y| \right]. \quad (2)$$

The simulations depict that 'pulling' the selected spot $O(O^X, O^Y)$ towards a misreporting agent's true preference is possible in the average method by simply moving the



(a) 1-Dimensional voting. (b) 2-Dimensional voting.

Fig. 8. Normalized cumulative agent utility deviation from the utility achieved using honest reporting, for different agent participation percentages as agents misreport, using mean and median voting rules.



(a) 1-Dimensional voting. (b) 2-Dimensional voting.

Fig. 9. Change of utility of a random agent while misreporting with respect to the deviation from the true preference, using mean and median voting rules.

preference in the opposite direction of that of the selected spot while misreporting, since the selected spot for this method is:

$$O^* = \frac{1}{n_A + n_B} \times \sum_{R \in \{A, B\}} \sum_{i=1}^{n_R} l_{R_i}^*. \quad (3)$$

where O^* represents the selected coordinates along any axis. It is observed that this has no effect on the median method since the selected spot for this method is computed as:

$$O^* = \begin{cases} l_{R(\frac{n+1}{2})}^*, & \text{if } n \text{ is even,} \\ l_{R(\frac{n}{2})}^*, & \text{otherwise.} \end{cases} \quad (4)$$

Thus, misreporting an agent's preferences in the opposite direction of the selected spot will not have a 'pulling' effect as the median remains unchanged. However, moving the selected spot away from an agent's true preferences is possible in both methods. Since this reduces the utility of the misreporting agent, the agent has no incentive to do so. In Fig. 8, the normalized cumulative utility deviations from the one using honest preferences have been presented, where the percentage of misreporting agents and the total number of agents vary. The utility deviates from the honest value regardless of the percentage of agents that misreport. The relation between an agent's preference deviation and the same agent's utility deviation is presented in Fig. 9 along with Pearson's and Spearman's correlation coefficients. The change in utility stays at or below 0 for the median method, whereas the mean method may

increase or decrease the utility. This is in agreement with the predictions since the utility of an agent ε_{R_i} is:

$$u_{R_i} = \left[1 - \left| l_{R_i}^X - O^X \right| \right] + \left[1 - \left| l_{R_i}^Y - O^Y \right| \right]. \quad (5)$$

The utility is inversely proportional to the distance between the chosen spot and the submitted preference of the agent ($u_{R_i} \propto 1/|l_{R_i}^* - O^*|$). As observed in the simulation results, the average method results in a lower fluctuation in this distance than honest preference submissions. In contrast, the median method does now allow this type of fluctuation. This results in the agents' utilities increasing occasionally for the average method but not for the median method.

V. DISCUSSION

The following were established from the research above:

- The 2-dimensional voting scenario differs from the 1-dimensional scenario because it allows all voters to vote in the same 2-dimensional space, thereby providing flexibility in preferences across both axes.
- The mean method of vote aggregation in both voting scenarios is not strategyproof, as agents' utilities may increase from misreporting.
- The median method of vote aggregation is strategyproof, capping the agent's utility from misreporting when reporting honestly.
- Using the median method, our proposed mechanism can aggregate votes in a strategyproof manner from agents living across an available space between two parallel or two pairs of mutually parallel roads.

VI. CONCLUSION

In conclusion, single-peaked preferences have proven to be a valuable tool for understanding the decision-making processes across various fields. This article extends the work on single-peaked preferences on a single line to single-peaked preferences on two parallel lines and two pairs of mutually parallel lines, and explores different methods for choosing the optimal point in the context of tourism. As technological platforms are available in smart cities, the paper is well-suited to the smart city context (although the mechanism can be applied in any context with a similar situation). The strategyproofness of the different methods used for the same has been analyzed and discussed using examples. The median method has been proven to be a truthful mechanism, where participants gain the most utility by honestly submitting their preferences to the system and lose when they misreport them.

It is worth noting that the model presented here assumes strict parallel lines and fixed preferences. In the future, this work will be extended by relaxing these assumptions to address even more general settings.

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